Quiz 3

BSSE23-A

Multivariable Calculus

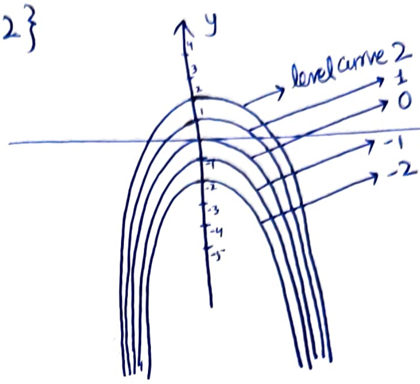
Q1: Sketch the level curve $z = k$ for the specified values of k

$z = x^2 + y$ for $k = \{-2, -1, 0, 1, 2\}$

Sol:

$x^2 + y = k$

OR $y = -x^2 + k$



Q2: Let $f(x,y) = \sin(3x^2 + 6y^2)$

Find f_{xx}, f_{yy}, f_{xy}, f_{yx}

Sol: $f_x = 6x\cos(3x^2+6y^2)$ and $f_y = 12y\cos(3x^2+6y^2)$

using Product rule.

So $f_{xx} = 6x[6x \cdot -\sin(3x^2+6y^2)] + \cos(3x^2+6y^2)\cdot 6$

$$\boxed{f_{xx} = 6\cos(3x^2+6y^2) - 36x^2\sin(3x^2+6y^2)}$$

using product rule.

and $f_{yy} = 12y[12y \cdot -\sin(3x^2+6y^2)] + \cos(3x^2+6y^2)\cdot 12$

$$\boxed{f_{yy} = 12\cos(3x^2+6y^2) - 144y^2\sin(3x^2+6y^2)}$$

and $f_{xy} = f_{yx} = 12y \cdot -\sin(3x^2+6y^2)\cdot 6x$

so $\boxed{f_{xy} = f_{yx} = -72xy\sin(3x^2+6y^2)}$

Q3: Show that the ellipsoid $2x^2+3y^2+z^2 = 9$, and the sphere $x^2+y^2+z^2-6x-8y-8z+24=0$ have a common tangent plane at point $(1,1,2)$

Sol: let $f_1(x,y,z) = 2x^2+3y^2+z^2$ and $f_2(x,y,z) = x^2+y^2+z^2-6x-8y-8z+24$

then $\vec{n}_1 = \nabla f_1 \big|_{at(1,1,2)} = \langle 4x, 6y, 2z\rangle = \langle 4, 6, 4\rangle$

and $\vec{n}_2 = \nabla f_2 \big|_{at(1,1,2)} = \langle 2x-6, 2y-8, 2z-8\rangle = \langle -4, -6, -4\rangle$

From $\vec{n}_1$ & $\vec{n}_2$ we see that $\vec{n}_1 = -\vec{n}_2$, so $\vec{n}_1 \parallel \vec{n}_2$

Hence f_1 & f_2 have a common tangent plane.